

Coding & Solution

Date	5 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/chandrika7177?tab=repositories
Programming Language(s)	Python, JavaScript, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	User Login, Soil Data Input, Crop Prediction using ML, Report Generation
Pending / Incomplete Features	Weather API integration, Mobile app support (if any)
Setup / Run Instructions	1. Install Python. 2. Install required libraries (pip install -r requirements.txt). 3. Run python app.py. 4. Open the localhost URL in a browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Crop Recommendation System is developed using Python, Flask, and Machine Learning. The code is modular, readable, and follows good programming practices. It predicts suitable crops based on soil parameters and provides accurate recommendations to users.